Assignment 5

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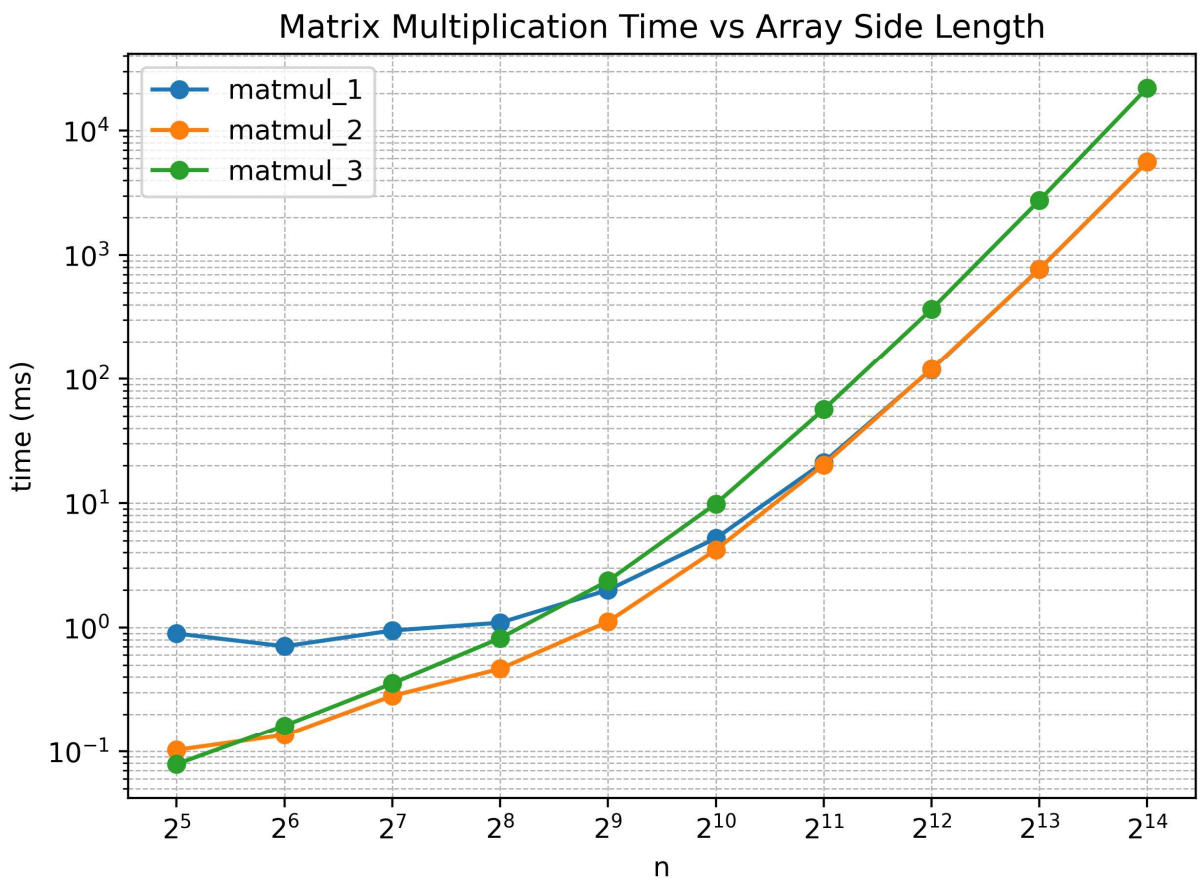
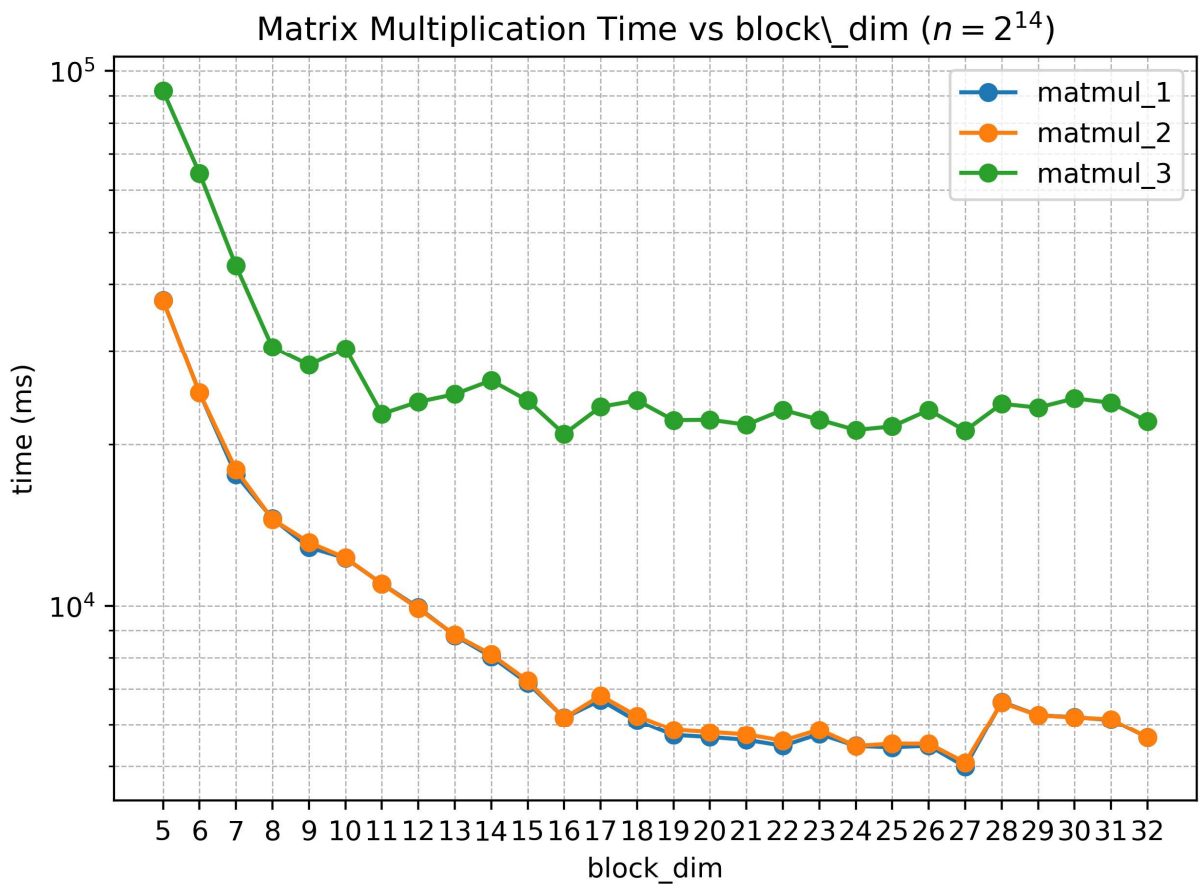
1) B)

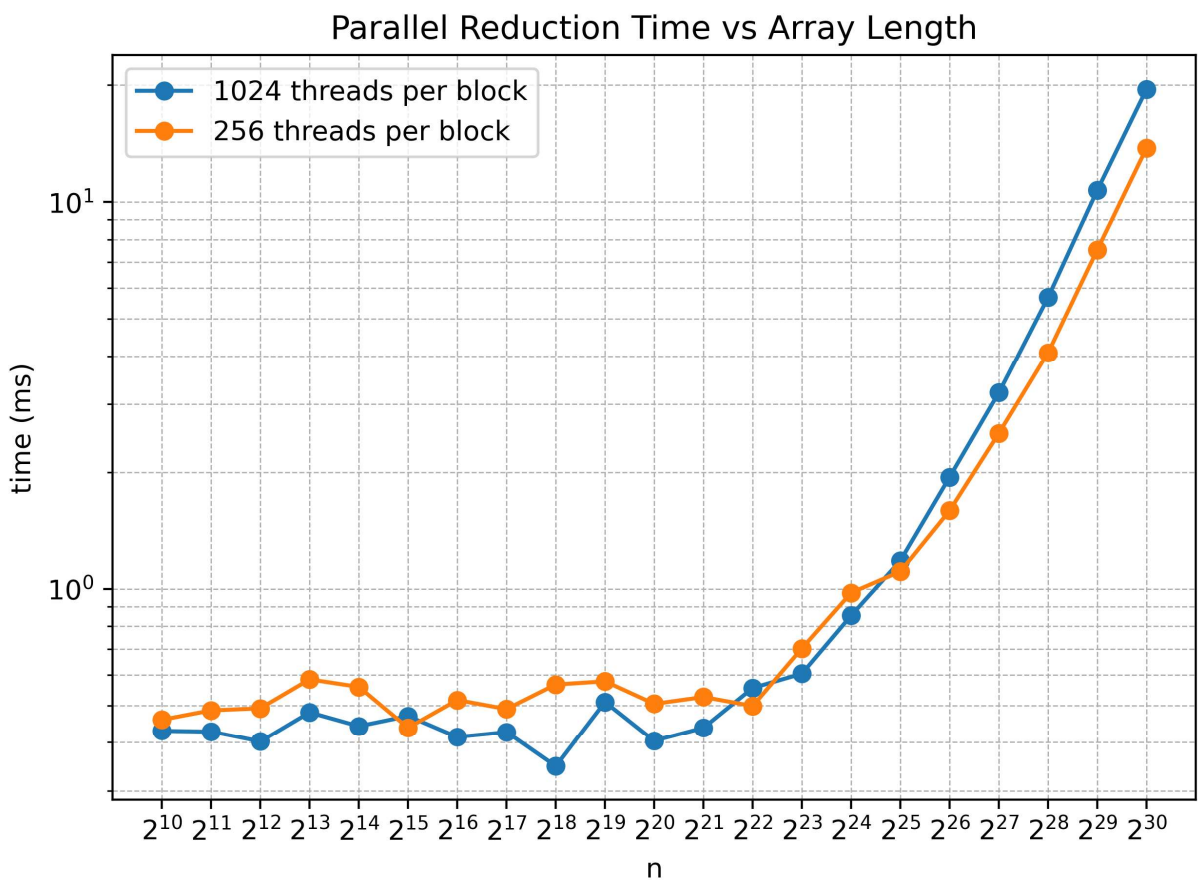


1) C) The best performing value of block_dim when n=2^{14} is **16** for matmul_1 and **27** for matmul_2 and matmul_3 as can be seen in the plot below:



1) D) The performance changes significantly depending on the type of data. This is probably because the data sizes are different, ints and floats are 4 bytes, doubles are 8 bytes. This application is mostly limited by memory traffic, and so the larger the data type, the slower the speed. This is confirmed by the fact that int and float perform almost the same speed and double performs less than half as well. The rest of the performance drop can be attributed to the fact the GPUs typically have lower FP64 throughput than FP32.

1) E) The best runtime for n=2^{14} in HW4 task 1 was 5949.62 ms, which compares with the best runtime for the float matmul (matmul_2) of 5080.91 ms at block dimension 27. The tiled version is thus faster because it utilizes shared memory, which is much faster than loading from global memory for every value in every thread.

1) F) The serial version went beyond 10 minutes, which is much slower than the runtime of the GPU implementation of around 5 seconds. This is because the GPU version takes advantage of massive parallelism. At $n = 2^{14}$ the matrix multiplication requires $\sim 2n^3 = 8.8 \times 10^{12}$ operations. On a single CPU core using a serial algorithm, this goes past 10 minutes, but since parallelism and memory can be taken advantage of much more effectively on a GPU, this GPU implementation is much faster.

2) B)